

Project 3

This week I worked on a 2,400-row E-commerce sales data in SSMS 22, writing SQL queries from scratch to extract real business insights.

Here's what I covered:

WHERE - I filtered orders by product, payment method, status & referral source.

ORDER BY I ranked orders by price and product category.

GROUP BY - I compiled 2,400 rows into groups of products, order status, payment methods & referral sources.

Aggregations (COUNT, SUM, AVG)

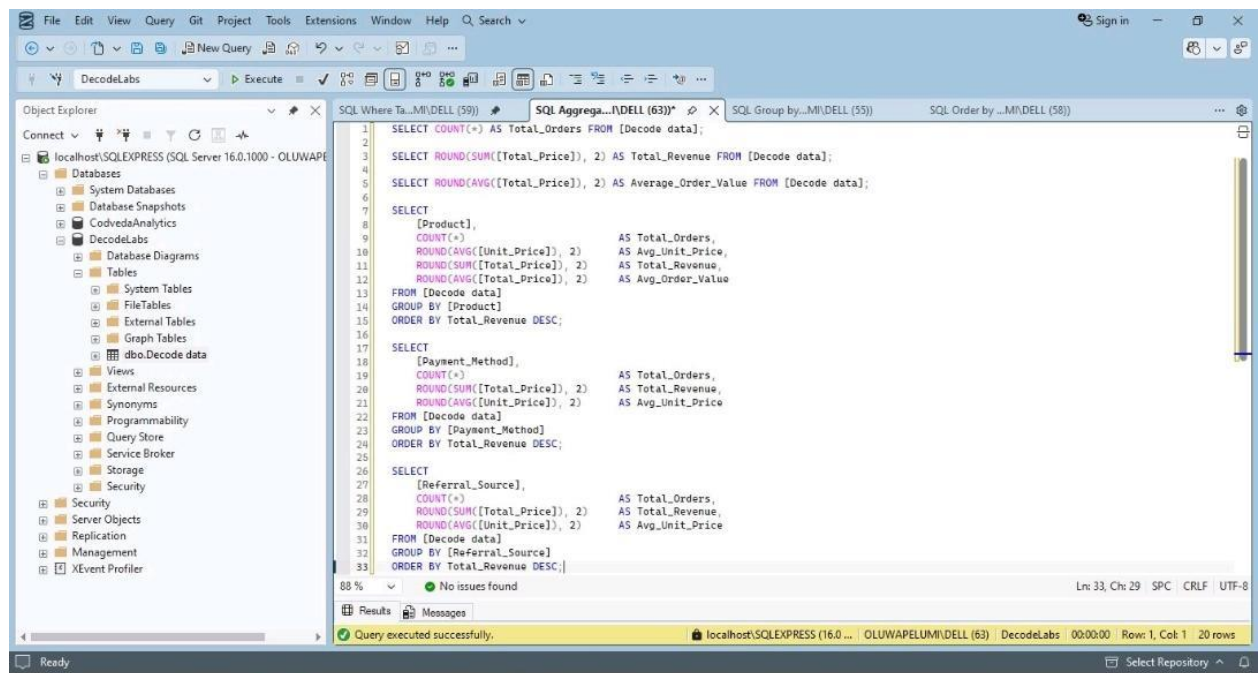
I calculated total revenue, average order value & per-product performance.

HAVING - I filtered grouped data to surface only high-performing categories.

Full Combined Query - Applied a query pulling total orders, units sold, revenue & averages for delivered products only.

Coupon Code Usage Analysis - I identified the most used discount codes across all orders

My Biggest lesson: SQL doesn't execute in the order you write it, mastering that execution order is everything.



SQL Server Enterprise Edition (64-bit) - SQL Server 16.0.1000 - OLUWAPE

Object Explorer: localhost\SQLEXPRESS (SQL Server 16.0.1000 - OLUWAPE) > Databases > DecodeLabs > Tables > dbo.Decode data

SQL Where Ta...MI\DELL (59) | SQL Havings ...I\DELL (52) | SQL Aggregat...I\DELL (63) | SQL Group by...MI\DELL (55)

```
1 SELECT [Product], COUNT(*) AS Total_Orders
2 FROM [Decode data]
3 GROUP BY [Product]
4 HAVING COUNT(*) > 50;
5
6 SELECT [Referral_Source], ROUND(SUM([Total_Price]), 2) AS Total_Revenue
7 FROM [Decode data]
8 GROUP BY [Referral_Source]
9 HAVING SUM([Total_Price]) > 50000;
10
11 SELECT [Payment_Method], ROUND(AVG([Total_Price]), 2) AS Avg_Order_Value
12 FROM [Decode data]
13 GROUP BY [Payment_Method]
14 HAVING AVG([Total_Price]) > 800;
```

88 % | No issues found | Ln: 14, Ch: 33 | TABS | CRLF | UTF-8

Results | Messages

Product	Total_Orders
Monitor	326
Phone	312
Tablet	358
Chair	356
Laptop	346
Desk	340
Printer	362

Referral_Source	Total_Revenue
Google	50082.96
Facebook	500821.8
Referral	453631.16
Email	523617.1
Instagram	550570.9

Query executed successfully. | localhost\SQLEXPRESS (16.0 ... OLUWAPE\MI\DELL (52) | DecodeLabs | 00:00:00 | Row: 1, Col: 1 | 7 rows

SQL Server Enterprise Edition (64-bit) - SQL Server 16.0.1000 - OLUWAPE

Object Explorer: localhost\SQLEXPRESS (SQL Server 16.0.1000 - OLUWAPE) > Databases > DecodeLabs > Tables > dbo.Decode data

SQLQuery8s...I\DELL (59) | SQLQuery9.sq...I\DELL (58) | Decode SQLQu...I\DELL (53)

```
1 SELECT * FROM [Decode data]
2 WHERE [Product] = 'Laptop';
3 SELECT [Order_ID], [Product], ROUND([Total_Price], 2) AS Total_Price
4
5 FROM [Decode data]
6 WHERE [Total_Price] >= 2000;
7
8 SELECT * FROM [Decode data]
9 WHERE [Payment_Method] = 'Credit Card';
10
11 SELECT [Order_ID], [Customer_ID], [Product], [Order_Status]
12 FROM [Decode data]
13 WHERE [Order_Status] = 'Delivered';
14
15 SELECT [Order_ID], [Product], ROUND([Total_Price], 2) AS Total_Price
16 FROM [Decode data]
17 WHERE [Product] = 'Laptop' AND [Order_Status] = 'Delivered';
18
19 SELECT [Order_ID], [Product], [Referral_Source], [Order_Status]
20 FROM [Decode data]
21 WHERE [Referral_Source] = 'Instagram' AND [Order_Status] = 'Shipped';
```

100 % | No issues found | Ln: 16, Ch: 19 | TABS

Results | Messages

Order_ID	Date	Customer_ID	Product	Quantity	Unit_Price	Shipping_Address	Payment_Method	Order_Status	Tracking_Number	Items_In_Cart
ORD200006	2025-06-17	C83492	Laptop	1	664.419982910156	986 Main St	Gift Card	Returned	TRK96417362	6
ORD200013	2023-08-30	C88348	Laptop	3	201.490005493164	903 Main St	Credit Card	Returned	TRK81913874	4
ORD200024	2024-10-30	C17470	Laptop	2	230.949996948242	956 Main St	Online	Delivered	TRK28775057	3
ORD200026	2024-03-06	C21944	Laptop	5	59.5499992370605	296 Main St	Online	Delivered	TRK80276812	6
ORD200036	2025-05-08	C59337	Laptop	2	534.429992675781	465 Main St	Debit Card	Pending	TRK50961250	5
ORD200037	2024-10-14	C10054	Laptop	2	76.7399978637695	409 Main St	Cash	Returned	TRK94005261	7

Query executed successfully. | localhost\SQLEXPRESS (16.0 ... OLUWAPE\MI\DELL (59) | DecodeLabs | 00:00:00 | Row: 1, Col: 1 | 7 rows

SQL Server Enterprise Edition (64-bit) - DecodeLabs

Object Explorer: localhost\SQLEXPRESS (SQL Server 16.0.1000 - OLUWAP...

SQL Where Ta...M\DELL (59) | SQL Order by ...M\DELL (58) | Decode SQLQ...DELL (53)*

```
1 SELECT [Product], COUNT(*) AS Total_Orders
2 FROM [Decode data]
3 GROUP BY [Product];
4
5 SELECT [Order_Status], COUNT(*) AS Total_Orders
6 FROM [Decode data]
7 GROUP BY [Order_Status];
8
9 SELECT [Payment_Method], COUNT(*) AS Total_Orders
10 FROM [Decode data]
11 GROUP BY [Payment_Method];
12
13 SELECT [Referral_Source], COUNT(*) AS Total_Orders
14 FROM [Decode data]
15 GROUP BY [Referral_Source];
```

100 % | No issues found | Ln: 15, Ch: 28 | TABS | CRLF | UTF-8

Results | Messages

	Product	Total_Orders
1	Monitor	326
2	Phone	312
3	Tablet	358
4	Chair	356
5	Laptop	346
6	Desk	340
7	Printer	362

	Order_Status	Total_Orders
1	Pending	474
2	Returned	494
3	Delivered	462

Query executed successfully. | localhost\SQLEXPRESS (16.0 ... | OLUWAPELUM\DELL (53) | DecodeLabs | 00:00:00 | Row: 1, Col: 1 | 22 rows

SQL Server Enterprise Edition (64-bit) - DecodeLabs

Object Explorer: localhost\SQLEXPRESS (SQL Server 16.0.1000 - OLUWAP...

SQL Where Ta...M\DELL (59) | SQL Having...M\DELL (52)* | SQL Aggregat...M\DELL (63)* | SQL Group by...M\DELL (55)

```
1 SELECT [Product], COUNT(*) AS Total_Orders
2 FROM [Decode data]
3 GROUP BY [Product]
4 HAVING COUNT(*) > 50;
5
6 SELECT [Referral_Source], ROUND(SUM([Total_Price]), 2) AS Total_Revenue
7 FROM [Decode data]
8 GROUP BY [Referral_Source]
9 HAVING SUM([Total_Price]) > 50000;
10
11 SELECT [Payment_Method], ROUND(AVG([Total_Price]), 2) AS Avg_Order_Value
12 FROM [Decode data]
13 GROUP BY [Payment_Method]
14 HAVING AVG([Total_Price]) > 800;
```

100 % | No issues found | Ln: 1, Ch: 1 | TABS | CRLF | UTF-8

Results | Messages

	Product	Total_Orders
1	Monitor	326
2	Phone	312
3	Tablet	358
4	Chair	356

Query executed successfully. | localhost\SQLEXPRESS (16.0 ... | OLUWAPELUM\DELL (52) | DecodeLabs | 00:00:00 | Row: 1, Col: 1 | 17 rows

SQL Server Enterprise Edition (64-bit) - DecodeLabs

Object Explorer: localhost\SQLEXPRESS (SQL Server 16.0.1000 - OLUWAPELUMI\DELL (57))

SQL Where Ta...MI\DELL (59)

```
1 SELECT
2     [Product],
3     COUNT(*) AS Total_Orders,
4     SUM([Quantity]) AS Total_Units_Sold,
5     ROUND(AVG([Unit_Price]), 2) AS Avg_Unit_Price,
6     ROUND(SUM([Total_Price]), 2) AS Total_Revenue,
7     ROUND(AVG([Total_Price]), 2) AS Avg_Order_Value
8 FROM [Decode data]
9 WHERE [Order_Status] = 'Delivered'
10 GROUP BY [Product]
11 HAVING COUNT(*) >= 5
12 ORDER BY Total_Revenue DESC;
```

Results: 7 rows

Product	Total_Orders	Total_Units_Sold	Avg_Unit_Price	Total_Revenue	Avg_Order_Value
Laptop	80	242	334.25	81428.86	1017.86
Phone	76	202	407.26	80690.82	1061.72
Printer	58	176	414.16	76109.46	1312.23
Monitor	62	192	371.55	71999.24	1161.28
Tablet	56	170	368.02	63589.04	1135.52
Chair	66	210	302.46	62931.66	953.51
Desk	64	188	276.64	48451.56	757.06

Query executed successfully. localhost\SQLEXPRESS (16.0.1000 - OLUWAPELUMI\DELL (57)) DecodeLabs 00:00:00 Rows: 1, Cols: 1 | 7 rows

SQL Server Enterprise Edition (64-bit) - DecodeLabs

Object Explorer: localhost\SQLEXPRESS (SQL Server 16.0.1000 - OLUWAPELUMI\DELL (54))

SQL Coupon ...I\DELL (54)*

```
1 SELECT [Coupon_Code], COUNT(*) AS Orders_Used
2 FROM [Decode data]
3 WHERE [Coupon_Code] IS NOT NULL AND [Coupon_Code] <> ''
4 GROUP BY [Coupon_Code]
5 ORDER BY Orders_Used DESC;
```

Results: 4 rows

Coupon_Code	Orders_Used
FREESHIP	626
No Coupon	618
WINTER15	584
SAVE10	572

Query executed successfully. localhost\SQLEXPRESS (16.0.1000 - OLUWAPELUMI\DELL (54)) DecodeLabs 00:00:00 Rows: 1, Cols: 1 | 4 rows